15. **Why might modular design benefit a system extended from a prototype?**

A) It increases vendor lock-in

B) It allows independent updates to components (e.g., analytics, reporting)

C) It prioritizes rapid deployment over security

D) It eliminates stakeholder involvement

16. **Software that excludes users with disabilities by ignoring screen-reader compatibility violates which ethical principle?**

A) Environmental Impact

B) Lack of Accessibility

C) Misinformation Spread

D) Surveillance & Exploitation

17. **Which ethical concern relates to energy-inefficient code contributing to high carbon emissions?**

A) Job Displacement

B) Security Negligence

C) Environmental Impact

D) Lack of Transparency

18. **Releasing software with known vulnerabilities without patching them is an example of:**

A) Intellectual Property Theft

B) Security Negligence

C) Algorithmic Bias

D) Privacy Violations

19. **Which ethical issue involves "black box" AI systems that obscure decision-making processes?**

A) Lack of Transparency

B) Job Displacement

C) Surveillance & Exploitation

D) Environmental Impact

17. Reengineering is the preferred option when a legacy system:
A. Has low business value
B. Is poorly documented
C. Uses obsolete hardware
D. Is business-critical but difficult to maintain

---

18. Compared with complete system replacement, software reengineering offers:
A. New functionality
B. Faster innovation
C. Lower technical and business risk
D. Higher long-term cost

---

19. Beta testing is particularly useful for identifying:
A. Syntax and compilation errors
B. Internal logic defects
C. Design pattern violations
D. Problems arising from real-world usage

---

20. Acceptance testing is successful when customers confirm that the system:
A. Follows coding standards
B. Is fully documented
C. Meets their operational and business needs
D. Has passed all developer tests

FINAL EXAM SPRING 2025
SOFTWARE ENGINEERING (Comp 303)

"Assume you are developing an online system for managing bank accounts for mobile customers. A major design issue is how to provide access to the accounts when the customer cannot establish an online connection. One proposal is that accounts are made available on the mobile computer, even if the server is not up. In this case, the accounts show the amounts from the last connected session."

30. **In the statement:**

"Assume you are developing an online system for managing `bank accounts` for mobile customers," the term "accounts" refers to a concept in the:

A) Application Domain

B) Solution Domain

C) Both Domains

D) Neither Domain

31. **The statement:**

"A major design issue is how to provide access to the `accounts` when the customer cannot establish an online connection"

uses the term "accounts" in the context of the:

A) Application Domain
B) Solution Domain
C) Both Domains
D) Neither Domain

32. **In the proposal:**

"The `accounts` show the amounts from the last connected session,"

the term "accounts" is part of the:

A) Application Domain
B) Solution Domain
C) Both Domains
D) Neither Domain

33. **Which statement best distinguishes the application domain use of "accounts"?**

A) "Accounts are stored locally on the mobile device."
B) "Accounts represent financial entities managed by the bank."
C) "Accounts synchronize with the server when online."
D) "Accounts use encryption for security."

34. **The phrase** *"accounts are made available on the mobile computer, even if the server is not up"* **primarily addresses:**

A) A business requirement (application domain).
B) A technical design choice (solution domain).

C) To replace real-world testing entirely
D) To eliminate stakeholder collaboration

44. **Which purpose of modeling is demonstrated by using UML diagrams to align developers and clients?**
A) Simplifying complexity
B) Facilitating communication
C) Supporting decision-making
D) Validating requirements

45. **Which purpose of modeling is achieved by prototyping a software interface before development?**
A) Simplifying complexity
B) Validating requirements
C) Guiding design
D) Enabling analysis

46. **Statement**: *"The system must be usable."*
This nonfunctional requirement is:
A) Verifiable
B) Non-Verifiable

47. **Statement**: *"The system must provide visual feedback to the user within one second of issuing a command."*
This nonfunctional requirement is:
A) Verifiable
B) Non-Verifiable

48. **Statement**: *"The availability of the system must be above 95 percent."*
This nonfunctional requirement is:
A) Verifiable
B) Non-Verifiable

49. **Statement**: *"The user interface of the new system should be similar enough to the old system that users familiar with the old system can be easily trained to use the new system."*
This nonfunctional requirement is:
A) Verifiable
B) Non-Verifiable

50. **Which architectural pattern organizes the drone's onboard software into modular layers (e.g., hardware abstraction, control, application)?**
A) Client-Server Architecture
B) Layered Architecture
C) Event-Driven Architecture
D) Microservices Architecture

C) Both domains equally.
D) A regulatory compliance issue.

35. **Statement:** *"The TicketDistributor must enable a traveler to buy weekly passes."*
This requirement is best classified as a:
A) Functional Requirement
B) Nonfunctional Requirement

36. **Statement:** *"The TicketDistributor must be written in Java."*
This requirement falls under:
A) Functional Requirement
B) Nonfunctional Requirement

37. **Statement:** *"The TicketDistributor must be easy to use."*
This is an example of a:
A) Functional Requirement
B) Nonfunctional Requirement

38. **Statement:** *"The TicketDistributor must always be available."*
This requirement is categorized as:
A) Functional Requirement
B) Nonfunctional Requirement

39. **Statement:** *"The TicketDistributor must provide a phone number to call when it fails."*
This is a:
A) Functional Requirement
B) Nonfunctional Requirement

40. **Statement:** *"The TicketDistributor is composed of a user interface subsystem, a subsystem for computing tariff, and a network subsystem for managing communication with the central computer."*
This decision is made during the:
A) Requirements Phase
B) System Design Phase

41. **Statement:** *"The TicketDistributor hardware uses PowerPC processor chips."*
This choice is finalized in the:
A) Requirements Phase
B) System Design Phase

42. **Statement:** *"The TicketDistributor provides the traveler with online help."*
This requirement is defined in the:
A) Requirements Phase
B) System Design Phase

43. **What is the primary purpose of modeling in system design?**
A) To increase system complexity for better functionality
B) To simplify complex systems by focusing on essential aspects

A) User interface design

B) Security and compliance measures

C) Basic data entry functionality

D) Stakeholder presentations

11. **Which strategy mitigates the risk of "scope creep" when extending a prototype?**

A) Avoiding stakeholder feedback

B) Formalizing requirements and modular design

C) Reusing all prototype code without changes

D) Delivering updates annually

12. **Why might a government face "vendor lock-in" when extending a prototype?**

A) The prototype uses open-source tools

B) The original developer retains exclusive control over updates

C) Multiple vendors collaborate on the system

D) The system is fully documented

13. **What is a potential consequence of incomplete requirements when scaling a prototype?**

A) Faster deployment

B) Missing critical features (e.g., compliance reporting)

C) Reduced technical debt

D) Lower stakeholder engagement

14. **Which scenario BEST illustrates "scope creep" in this context?**

A) Delaying the project to fix security flaws

B) Adding AI-driven mineral price forecasting mid-development

C) Reducing features to meet deadlines

D) Conducting third-party audits

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QUESTION 1: CHOOSE THE CORRECT CHOICE FOR EACH OF THE FOLLOWING [1 MARK/EACH]

Consider the following statement of the requirements for part of a drone system intended for search and recovery:

The drone, a quadcopter, will be very useful in search and recovery operations, especially in remote areas or in extreme weather conditions. It will click high-resolution images. It will fly according to a path preset by a ground operator but will be able to avoid obstacles on its own, returning to its original path whenever possible. The drone will also be able to identify various objects and match them to the target it is looking for.

1. What is a key ambiguity in the requirement stating the drone will "click high-resolution images"?

A) The type of camera used (e.g., thermal, RGB)
B) The lack of a defined resolution standard (e.g., 4K, 1080p)
C) How frequently images are transmitted to the operator
D) The storage format for the images

2. Which critical detail is omitted about the drone's preset flight path?

A) How the ground operator defines or inputs the path
B) The maximum altitude the drone can reach
C) The drone's ability to recalibrate mid-flight
D) The battery life during path execution

3. What is ambiguous about the drone's ability to "avoid obstacles on its own"?

A) The sensors or technology used for obstacle detection (e.g., LiDAR, cameras)
B) The maximum speed of the drone during avoidance
C) The altitude at which obstacles are detected
D) The color of obstacles the drone recognizes

4. What is unclear about the drone "returning to its original path whenever possible"?

A) The criteria for determining when it is "possible" to return
B) The drone's battery consumption during path deviation
C) The operator's role in manual path correction
D) The drone's maximum deviation distance

5. What is omitted about the drone's ability to "identify various objects and match them to the target"?

A) The types of objects the drone can recognize (e.g., humans, vehicles)
B) The algorithm or criteria used for matching objects to targets

5. A reuse-oriented approach may fail to meet real user needs primarily because:
A. Reusable components lack documentation
B. Integration frameworks are immature
C. Requirements must adapt to available components
D. Testing becomes complex

6. A system passes all functional tests but users reject it. Which activity was insufficient?
A. Verification
B. Validation
C. Integration testing
D. Unit testing

7. Why does prototyping reduce the risk of requirements volatility?
A. It eliminates formal specification
B. It replaces system testing
C. It delays final implementation
D. It enables early user feedback

8. Which Agile value most directly conflicts with traditional contract-driven development?
A. Individuals over processes
B. Customer collaboration
C. Responding to change
D. Working software

9. Refactoring is most effective when it is:
A. Performed at project milestones
B. Automated by tools
C. Conducted after testing
D. Continuous and incremental

10. A system meets all functional requirements but violates a timing constraint. What is the outcome?
A. Partial acceptance
B. Reduced efficiency
C. Minor defect
D. System failure

20. **Violating open-source licenses by plagiarizing code is an example of:**

A) Intellectual Property Theft
B) Environmental Impact
C) Lack of Accessibility
D) Algorithmic Bias

21. **Which ethical issue is highlighted by platforms amplifying extremist content through recommendation algorithms?**

A) Job Displacement
B) Misinformation Spread
C) Security Negligence
D) Lack of Transparency

22. **Why is adaptability important for long-term cost savings?**

A) It locks systems into rigid architectures
B) It allows systems to evolve with changing needs without costly rewrites
C) It prioritizes short-term fixes over sustainable design
D) It increases dependency on outdated technologies

23. **Which practice mitigates the risk of project failures?**

A) Ignoring risk assessments
B) Systematic risk management (e.g., prototyping)
C) Avoiding iterative development
D) Delaying testing until deployment

24. **Why might users of generic software lack control over new features?**

A) Updates are driven by vendor priorities, not individual needs
B) Customization options are unlimited
C) Users can directly modify the source code
D) Generic software is always open-source

25. **Which trade-off do users of generic software face compared to custom solutions?**

11. Abstraction in system modeling primarily serves to:
A. Eliminate system detail
B. Reduce development cost
C. Highlight essential system properties
D. Replace implementation

---

12. Architectural decisions have the strongest impact on:
A. Coding speed
B. Team organization
C. Testing strategy
D. System quality attributes

---

13. In MVC, separating View from Model primarily improves:
A. Performance
B. Security
C. Maintainability
D. Reliability

---

14. The Repository architectural pattern reduces coupling between components by:
A. Allowing components to communicate directly
B. Enforcing layered access rules
C. Centralizing data management
D. Replicating data across subsystems

---

15. If a system satisfies user expectations but violates documented specifications, this indicates a failure of:
A. Validation
B. Acceptance testing
C. Verification
D. Unit testing

---

16. Test-Driven Development (TDD) improves design quality mainly because it:
A. Eliminates the need for debugging
B. Encourages small, testable units of code
C. Replaces formal specifications
D. Reduces the number of required tests

CAIRO UNIVERSITY	**Math. Department** Final Exam for **Comp303** students Third level / First semester - Fall 2025-2026	
Course Name: Software Engineering **Code:** Comp303 **Instructor:** Dr Hassan Ayad		**Total Mark:**60 Marks **Allowed Time:**2 hours **Date:** Jan. 11, 2025

Question 1 (choose the correct choice in the answers table on the last page) [20 marks]

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1. A company expects frequent regulatory and market changes. Which software attribute most directly mitigates long-term cost?
A. Efficiency
B. Maintainability
C. Reliability
D. Usability

---

2. When a purchasing organization can redefine system functionality during development, the software is best classified as:
A. Embedded software
B. Generic product
C. Customized product
D. Commercial off-the-shelf software

---

3. The most serious management risk of the Waterfall model arises from:
A. Excessive documentation
B. Late integration
C. Rigid phase boundaries
D. Early commitment to fixed requirements

---

4. Which project context most strongly favors incremental development over the Waterfall model?
A. Stable requirements with fixed contracts
B. Safety-critical embedded systems
C. Systems requiring regulatory certification
D. Systems with evolving or unclear requirements

A) Higher upfront costs but lower long-term flexibility

B) Lower upfront costs but limited specificity

C) Full control over scalability and integration

D) Guaranteed compliance with niche regulations

26. **How might generic software limit scalability for growing organizations?**

A) It automatically integrates with custom tools

B) It cannot adapt to unique scaling needs without workarounds

C) It requires no additional training

D) It is always cloud-based

27. **Why might a small business choose generic accounting software over a custom solution?**

A) To fully control feature development

B) To avoid vendor dependency

C) To benefit from lower upfront costs and general features

D) To ensure compliance with industry-specific regulations

28. **What is the disadvantage of generic software for organizations with evolving workflows?**

A) It guarantees seamless integration with future tools

B) It forces reliance on vendor-driven updates for new capabilities

C) It eliminates the need for employee training

D) It provides unlimited customization options

29. **Which scenario highlights the limitation of generic software?**

A) A hospital uses a custom patient management system for specialized workflows.

B) A startup uses Slack but needs additional tools for custom automation as it grows.

C) A school develops proprietary software to manage student data.

D) A retail chain builds an in-house inventory system.

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C) The image resolution required for identification
D) The drone's flight speed during identification

6. What operational parameter is missing for "search and recovery in remote areas"?

A) The drone's flight duration (battery life)
B) The communication range between drone and operator
C) Both A and B
D) The weight of objects the drone can carry

7. What is ambiguous about the drone's use in "extreme weather conditions"?

A) The definition of "extreme weather" (e.g., wind speed, rain)
B) The drone's maximum speed in such conditions
C) The operator's training for weather emergencies
D) The color visibility of the drone in storms

Imagine that a government wants a software program that helps to keep track of the utilization of the country's vast mineral resources. Although the requirements put forward by the government were not very clear, a software company was tasked with the development of a prototype. The government found the prototype impressive, and asked it be extended to be the actual system that would be used.

8. What is the primary advantage of extending a prototype into a full system for tracking mineral resources?

A) Guaranteed long-term scalability
B) Rapid deployment and early stakeholder feedback
C) Elimination of all technical debt
D) Full compliance with regulatory standards

9. Which risk is MOST likely when scaling a prototype into a national system without redesign?

A) Excessive documentation
B) Improved stakeholder engagement
C) Technical debt and scalability issues
D) Reduced development costs

10. What critical component is often neglected in prototypes but essential for a national system?